1. Clinical Question & Core Hypothesis

Clinical Question

EMT and Wnt/beta-catenin activation drive metastatic, radioresistant phenotypes, yet radiotherapy can potentiate immunotherapy; why do some immune-infiltrated tumors still fail to respond to combined RT+immunotherapy? 2) Tier 1 path used: the single-path mechanistic backbone linking EMT ↔ Wnt/beta-catenin ↔ metastatic spread and modulation of radiosensitivity; I built on the beta-catenin node (CTNNB1) as the pathway effector and the EMT program as the cellular state. 3) Supporting (Tier 2) evidence: RT's ability to increase tumor immunogenicity implies that the state of tumor-intrinsic signaling (Wnt/CTNNB1-driven EMT) may prevent RT-mediated immune priming or function of existing TILs, explaining discordant responses. 4) Endpoint choice rationale: I selected therapy_response (response to combined radiation + immunotherapy) because the mechanistic path centers on treatment interaction (RT-induced immunogenicity vs. EMT/Wnt-driven resistance), making a therapy-response proxy the most direct, testable clinical outcome. 5) Deliberate exclusions: I excluded bladder-specific agents (BCG) and tissue-specific assumptions (urothelium vs. breast) from the core hypothesis to keep focus on the generalizable mechanism (CTNNB1-driven EMT undermining RT+immunotherapy efficacy) and to comply with the limit of ≤2 core molecular variables; those factors are retained as secondary modifiers for downstream exploratory analyses?

Working Hypothesis

Activation of CTNNB1 drives an EMT program that creates an immune-excluded, immunosuppressive tumor microenvironment and modulates radiation-induced immunogenic effects, resulting in diminished efficacy of combined radiotherapy and immunotherapy. Specifically, tumors with high CTNNB1 activity and corresponding EMT signatures will exhibit reduced cross-priming and effector CD8+ T cell activity, lower MHC-I presentation, and altered chemokine landscapes that hamper TIL recruitment and function, thereby yielding inferior therapy_response to RT+IO compared with tumors with low CTNNB1 activity and EMT. This effect is mechanistically linked to reduced dendritic cell recruitment (e.g., BATF3-DC–supporting chemokines such as CCL4/CXCL9), increased immunosuppressive signaling, and potential radioresistance pathways, and is expected to be observable across solid tumor contexts independent of tissue origin when CTNNB1/EMT states are matched. Exploratory modifiers (e.g., tumor tissue context or BCG-related urothelial activation) may modulate the magnitude but not the direction of the effect.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CTNNB1, EMT_program
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CTNNB1: CTNNB1-driven EMT undermining RT+immunotherapy efficacy) and to comply with the limit of ≤2 core molecular variables; those factors are retained as secondary modifiers for downstream exploratory analyses.

EMT_program: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	Wnt/beta-catenin pathway activity	Supporting modifier from knowledge graph
Secondary	BCG-induced urothelial immune activation	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In vitro: manipulate CTNNB1 activity and EMT state in cancer cell lines; measure EMT markers, MHC-I, PD-L1, chemokines (e.g., CCL4, CXCL9), and susceptibility to CD8+ T cell–mediated killing;

Prediction 2: In vivo: use syngeneic mouse models with controllable CTNNB1 activity; apply focal RT with/without anti-PD-1 or anti-CTLA-4; quantify tumor growth, survival, and TIL density/phenotype;

Prediction 3: Mechanistic readouts: assess dendritic cell recruitment (BATF3 lineage), T cell priming, and cytokine milieu;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Wnt/beta-catenin pathway activity: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- BCG-induced urothelial immune activation: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.